

Organization of Digital Computers Lab

EECS 112L

Lab #5: Implementing the ALU design on Basys 3 FPGA

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1 Objective

In this lab experiment, we going to be measuring the time, power, and area for the pipeline processor. We will be using constraints for the synthesis and implementing the ALU design on Basys 3 FPGA.

Artix-7 FPGA Feature Summary

Table 4: Artix-7 FPGA Feature Summary by Device

Device	Logic Cells	Configurable Logic Blocks (CLBs)		DSP48E1 Slices ⁽²⁾	Block RAM Blocks ⁽³⁾			CMTs ⁽⁴⁾	PCOs ⁽⁵⁾	GTPs	XADC Blocks	Total I/O Banks ⁽⁶⁾	Max User I/O ⁽⁷⁾
		Slices ⁽¹⁾	Max Distributed RAM (Kb)		18 Kb	36 Kb	Max (Kb)						
XC7A12T	12,800	2,800	171	40	40	20	720	3	1	2	1	3	150
XC7A15T	16,640	2,800	200	45	50	25	800	5	1	4	1	5	250
XC7A25T	23,360	3,650	213	60	60	45	1,600	3	1	4	1	3	160
XC7A35T	33,280	5,200	400	80	100	50	1,800	5	1	4	1	5	250
XC7A60T	62,160	8,100	600	120	150	75	2,700	5	1	4	1	5	250
XC7A75T	75,320	11,600	692	160	210	105	3,700	5	1	5	1	6	300
XC7A100T	101,440	15,600	1,188	240	270	135	4,800	5	1	5	1	6	300
XC7A200T	210,560	33,600	2,688	740	720	360	13,140	10	1	10	1	10	500

Notes:

- Each 7 series FPGA slice contains four LUTs and eight flip-flops; only some slices can use their LUTs as distributed RAM or SRLs.
- Each DSP slice contains a pre-adder, a 25 x 18 multiplier, an adder, and an accumulator.
- Block RAMs are fundamentally 36 Kb in size; each block can also be used as two independent 18 Kb blocks.
- Each CMT contains one MMCM and one PLL.
- Artix-7 FPGAs interface blocks for PCI Express support up to x4 Gen 2.
- Does not include configuration flash I/O.
- This number does not include GTP transceivers.

Figure 1: Artix-7 family overview

2 Procedure

First, we are to create a new project using the FPGA device under parts: xc7a35tcpg236-1. Then, we set up our timing constraints. As stated in the lab manual, we are given the counter design and are to implement it and then provide the timing constraints summary. We will report the utilization summary and power summary as well. Furthermore, we are to report the schematic and provide commentary per component.

For the pipelined processor, we are to implement the MIPS pipelined processor. This was done in the previous lab. Like with the counter design, we return the timing constraints, utilization, and power summary along with giving a detailed summary about the schematic.

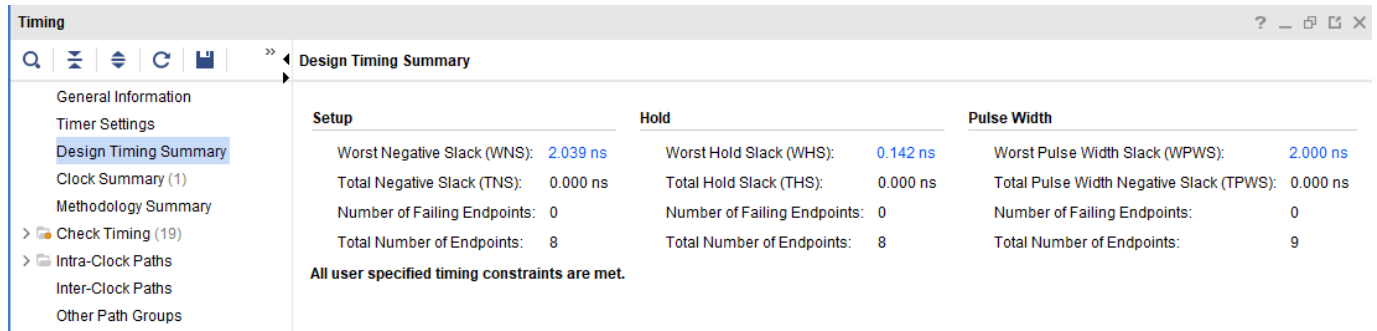
The second half portion of this lab is to implement the 32-bit ALU and connect the FPGA device, the Basys 3. We are given a truth table that serve as our test cases as seen below:

Case	A	B	ALU Sel	Output	Zero	Carry out	Overflow
i.	111	001	0000	001	0	0	0
ii.	100	000	0000	000	1	0	0
iii.	110	101	0001	111	0	0	0
iv.	000	000	0001	000	1	0	0
v.	001	010	0010	011	0	0	0
vi.	100	111	0010	011	0	1	1
vii.	110	110	0010	100	0	1	0
viii.	011	010	0110	001	0	0	0
ix.	100	111	0110	101	0	0	0
x.	001	011	0111	001	0	0	0
xi.	100	000	0111	001	0	0	0
xii.	111	001	1100	000	1	0	0
xiii.	001	101	1100	010	0	0	0
xiv.	000	000	1111	001	0	0	0
xv.	100	001	1111	000	1	0	0

3 Simulation Results

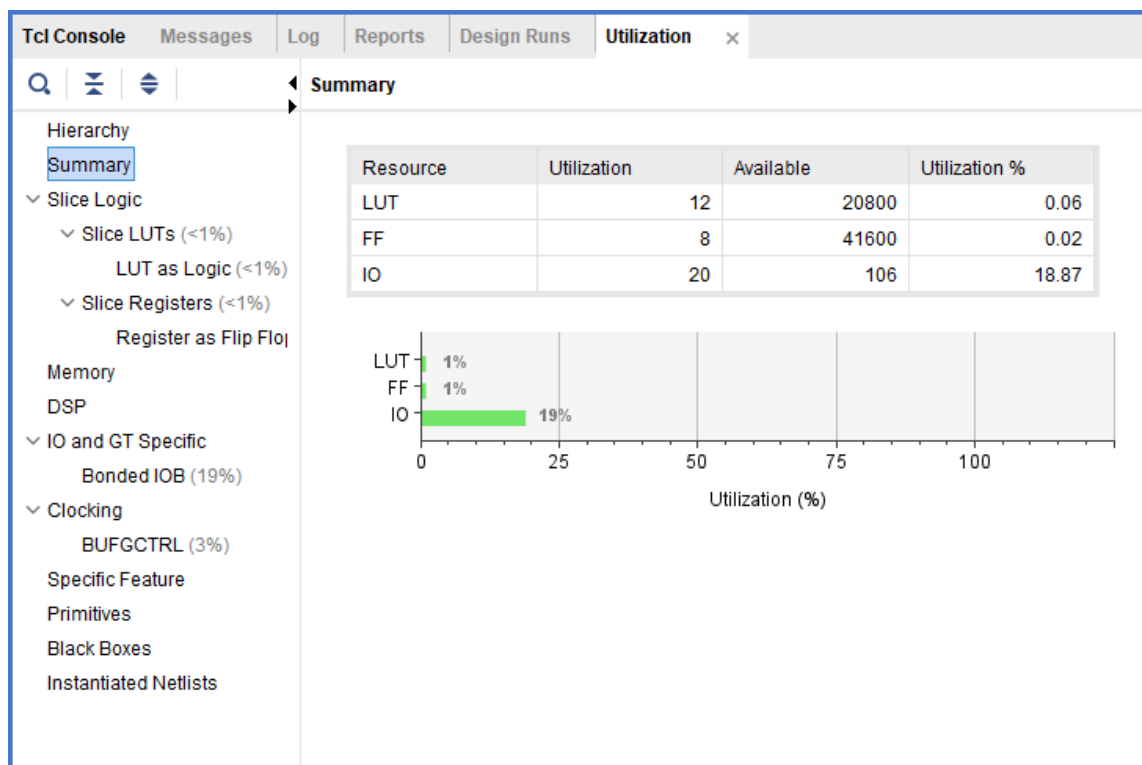
Synthesis

Timing Constraints:



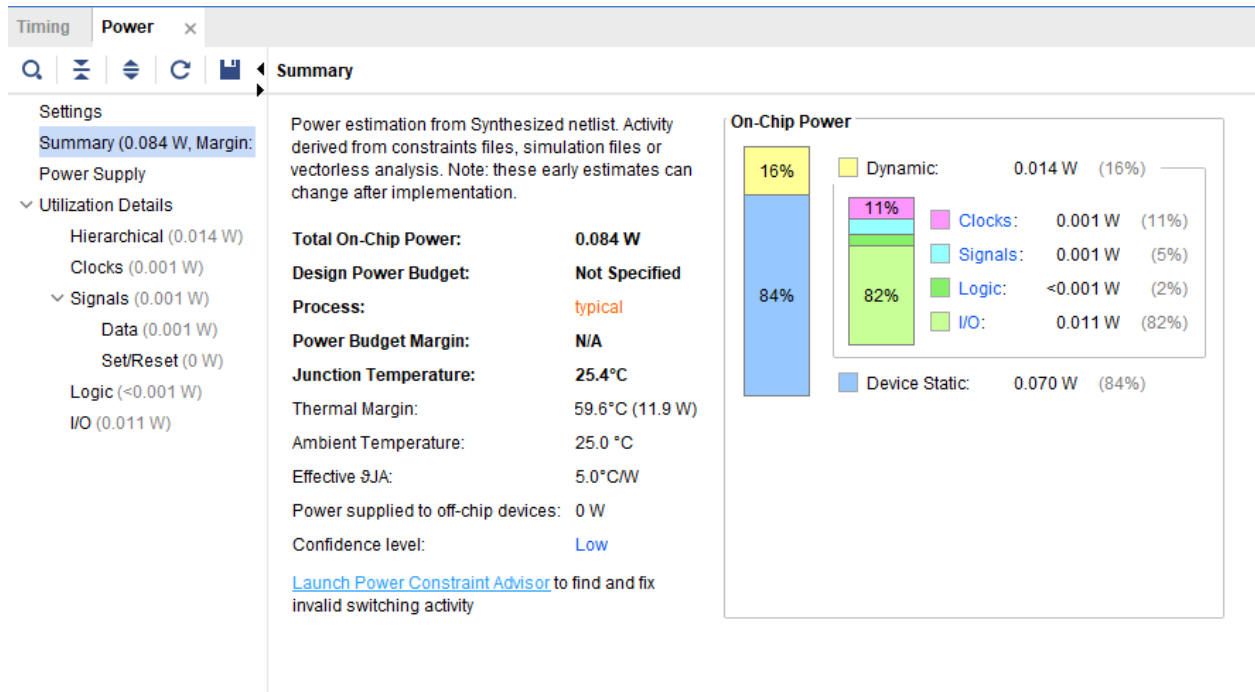
The timing was met by synthesis as the screenshot states “All user specified timing constraints are met.”

Utilization:



12 LUTs, 8 FFs, and 20 IOs were used by the design. From this resource consumption, 5 counters can fit on the FPGA since from the IO there are 106 available with 20 per counter thus 5.

Power:

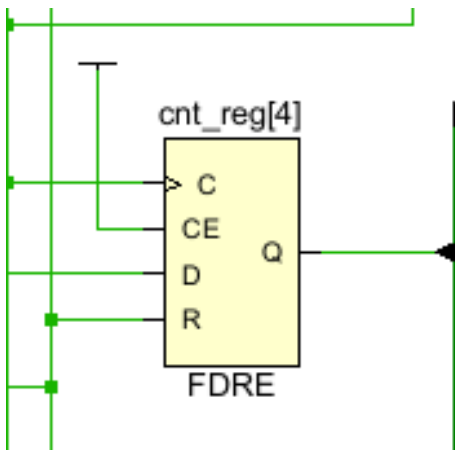
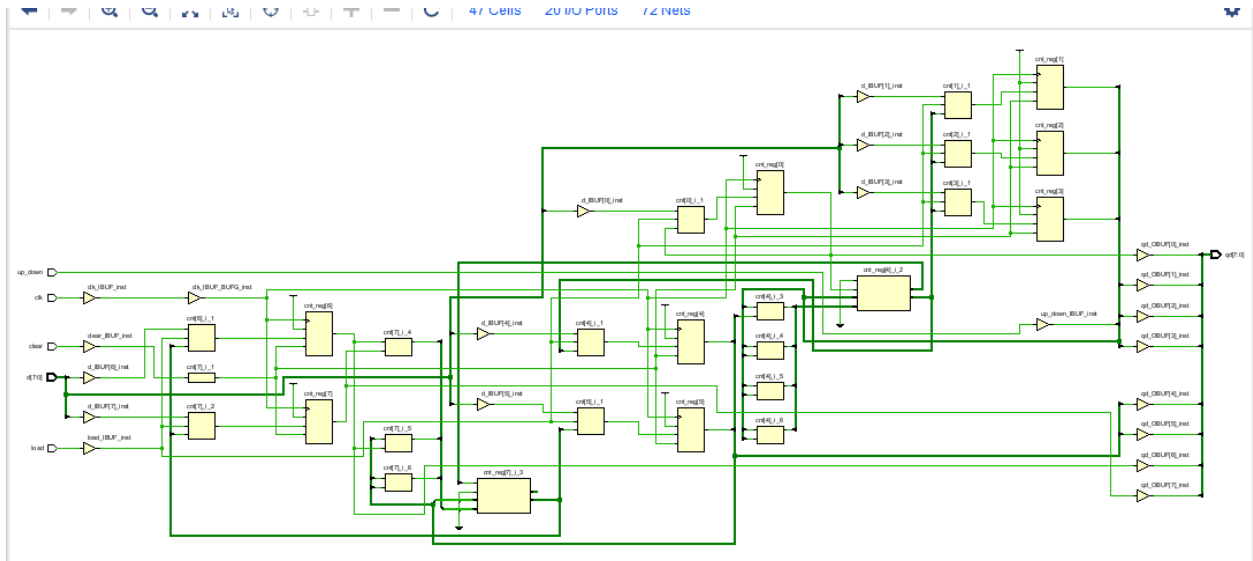


The Total On-Chip Power is 0.084 W.

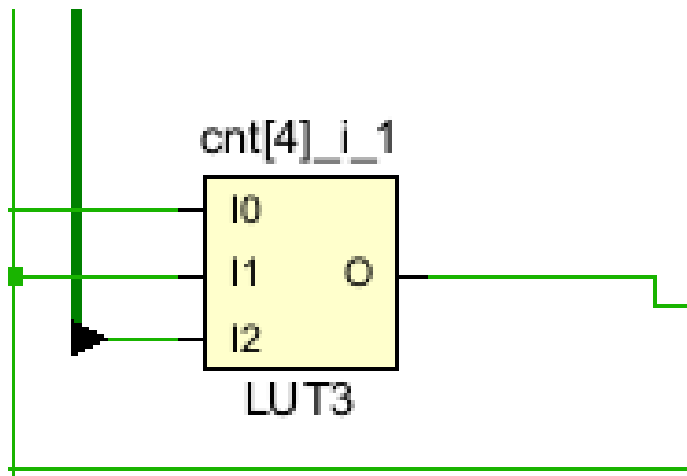
The Dynamic Power is 0.014 W, and the Static Power is 0.070 W. The Static Power is greater likely due to leakage current.

The greatest source of Dynamic Power consumption is the I/O because it contributes to 84% of dynamic power consumption. This occurs because of the switching rate of transistors, resulting in dynamic power, and this becomes the most prominent source of contributing.

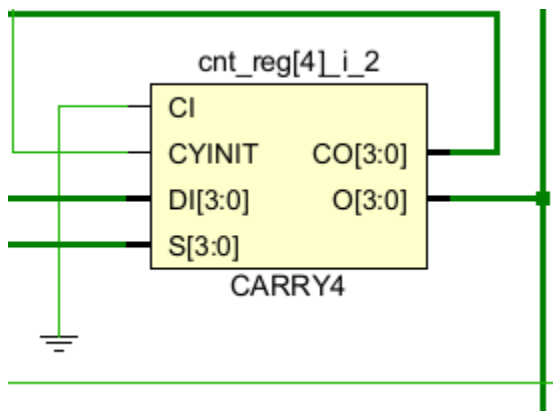
Schematic:



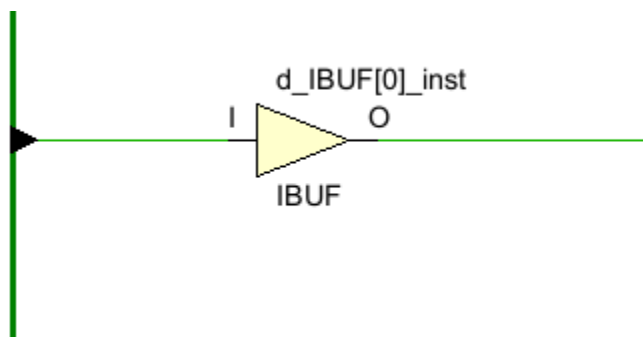
The above component is the FDRE, which is a D Flip Flop that has a reset signal that stores data at the rising edge of the clock.



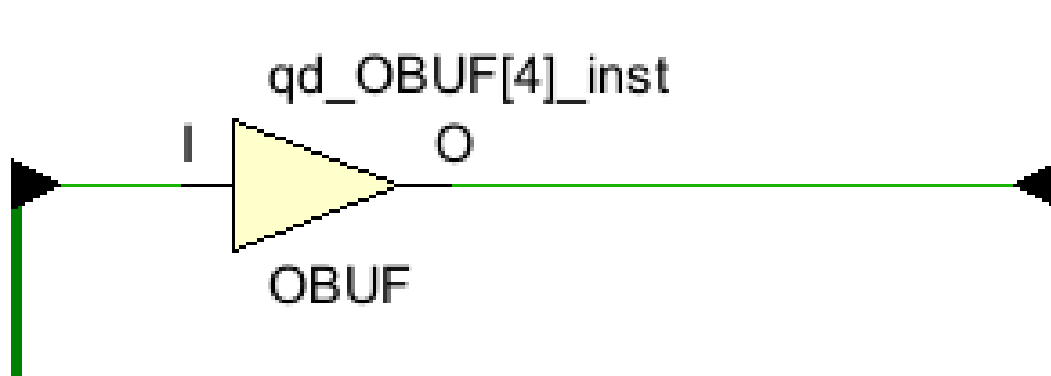
The LUT as seen above is used for color display taking in usually RGB and then synthesizes an output signal.



The CARRY4 as seen above is used for the carry necessities during arithmetic operations are executed.



The IBUF as seen above is used for being a buffer for receiving signals and provides a proper (or clean) signal to the internal logic of the chip.



The OBUF as seen above is used for handling signals from the internal logic of the chip to output pins.

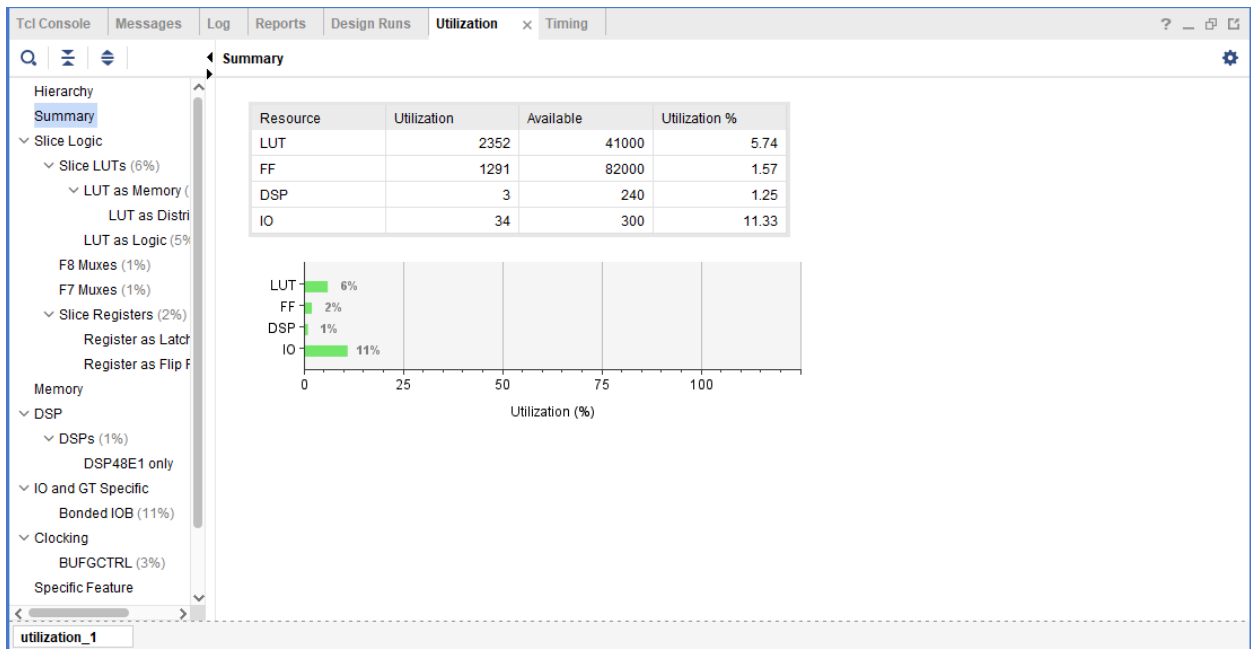
MIPS Pipelined Processor

Timing Constraints

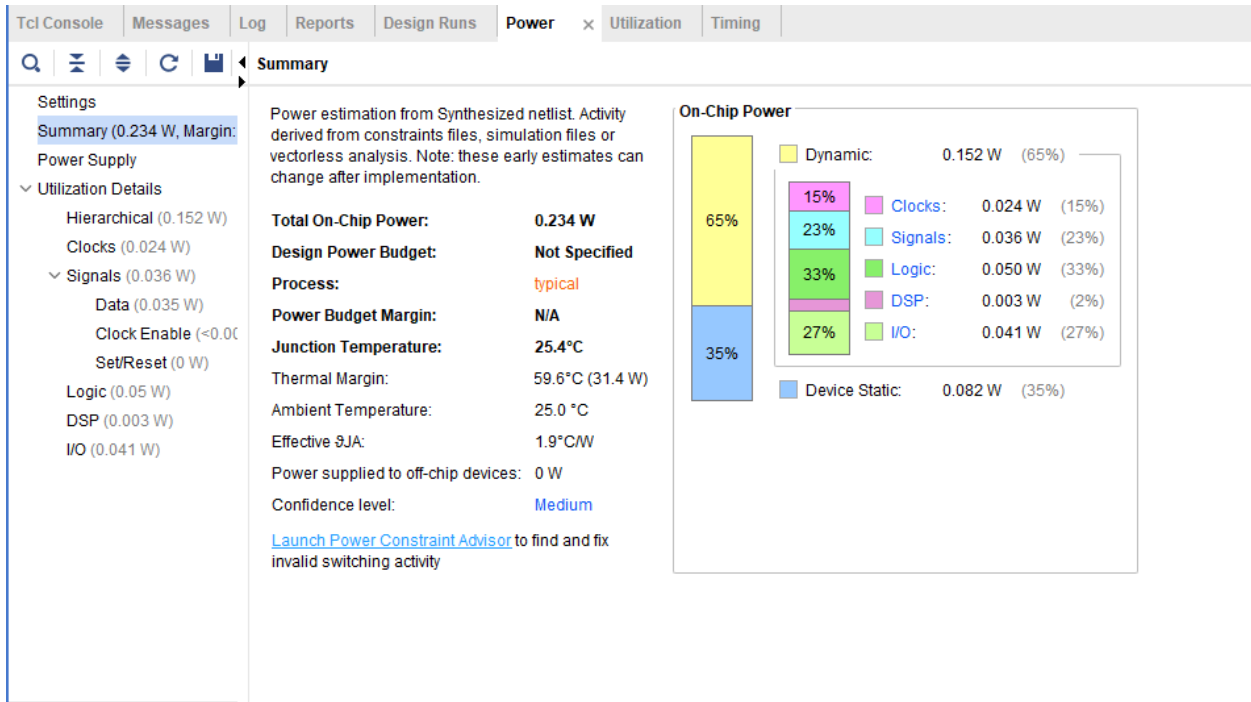
Design Timing Summary			
Setup	Hold	Pulse Width	
Worst Negative Slack (WNS): -43.962 ns	Worst Hold Slack (WHS): -0.025 ns	Worst Pulse Width Slack (WPWS): 1.590 ns	
Total Negative Slack (TNS): -937.395 ns	Total Hold Slack (THS): -10.964 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns	
Number of Failing Endpoints: 129	Number of Failing Endpoints: 768	Number of Failing Endpoints: 0	
Total Number of Endpoints: 3601	Total Number of Endpoints: 3601	Total Number of Endpoints: 1411	
Timing constraints are not met.			

The timing was not met by synthesis as the screenshot states “Timing constraints are met.”

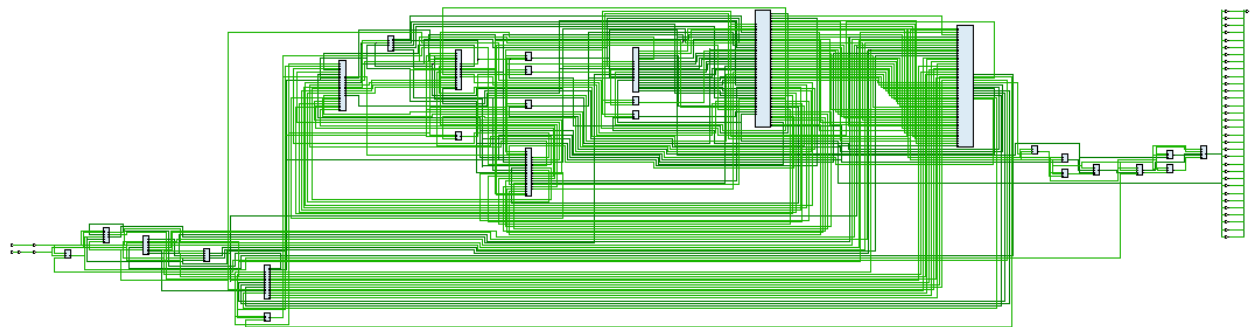
Utilization



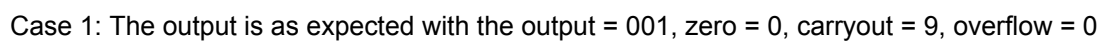
Power

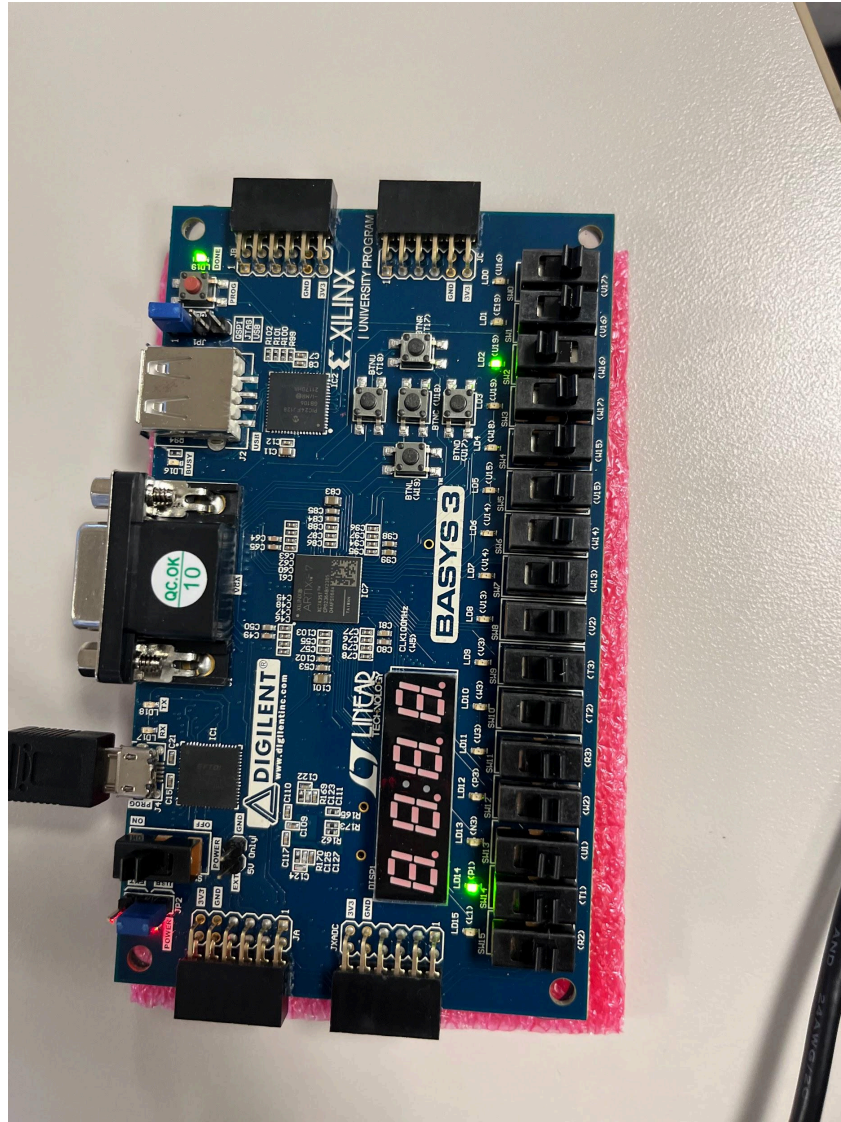


Schematic

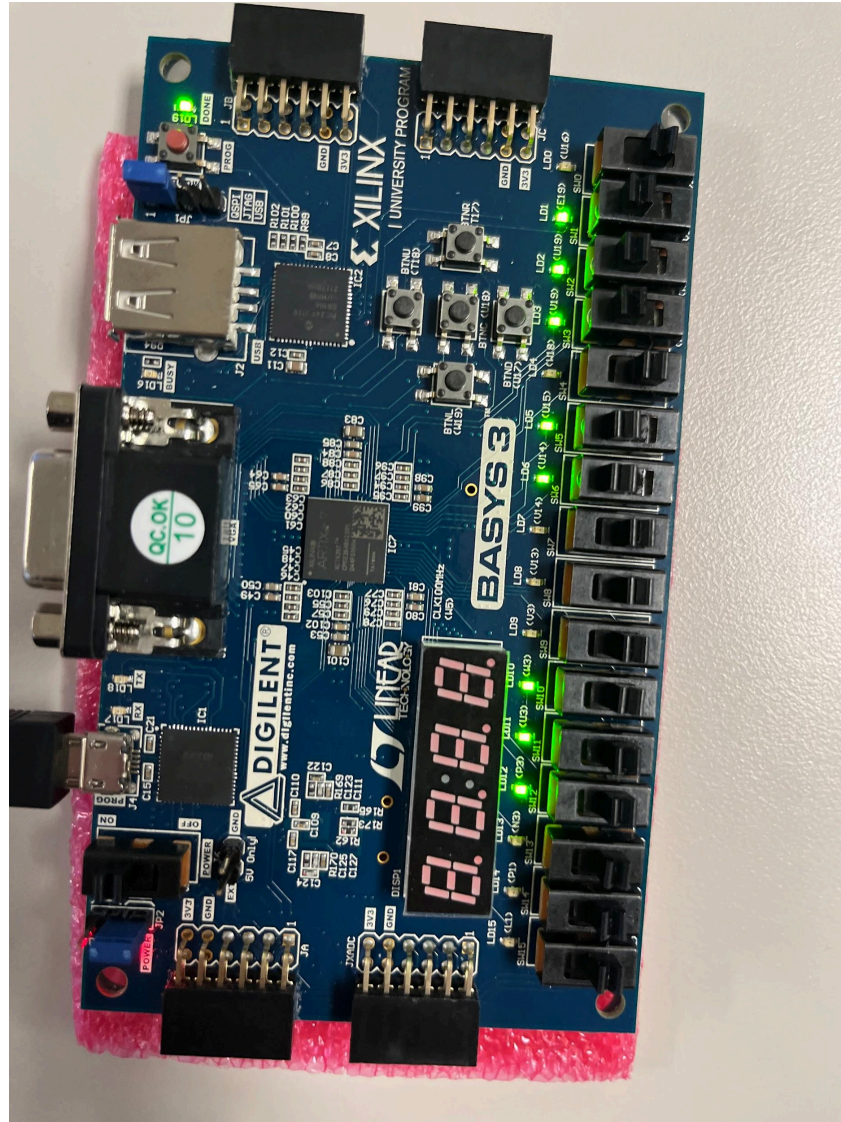


FPGA Results:

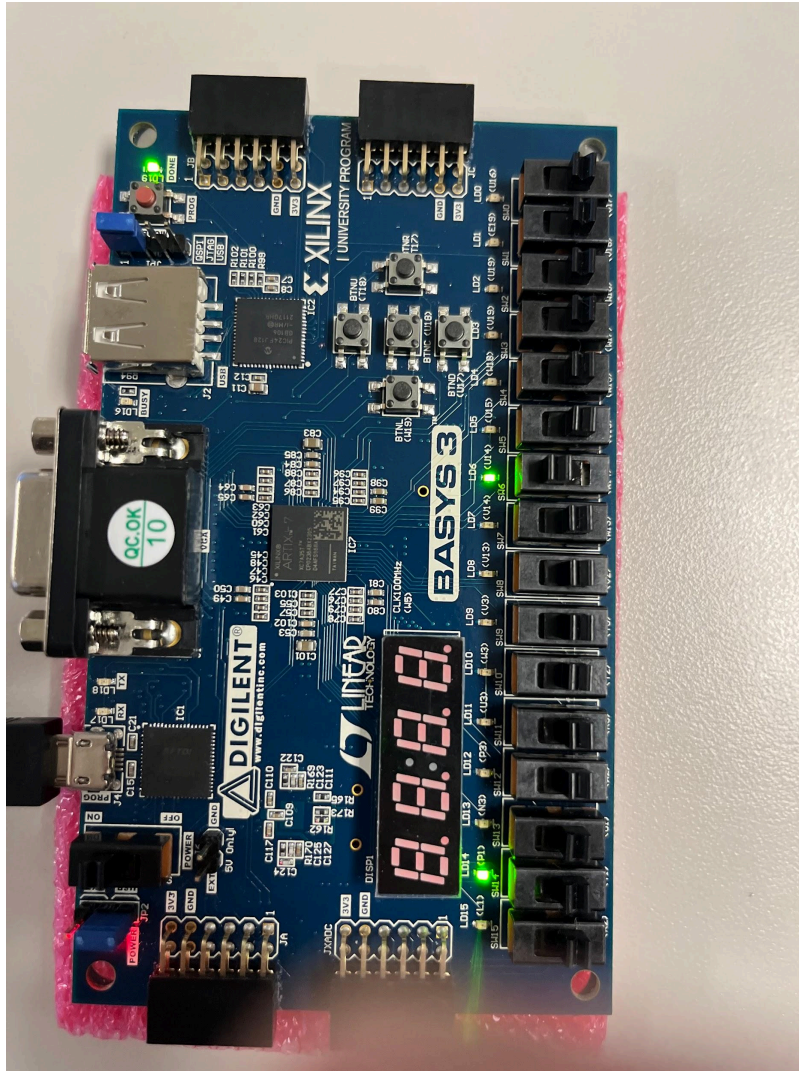




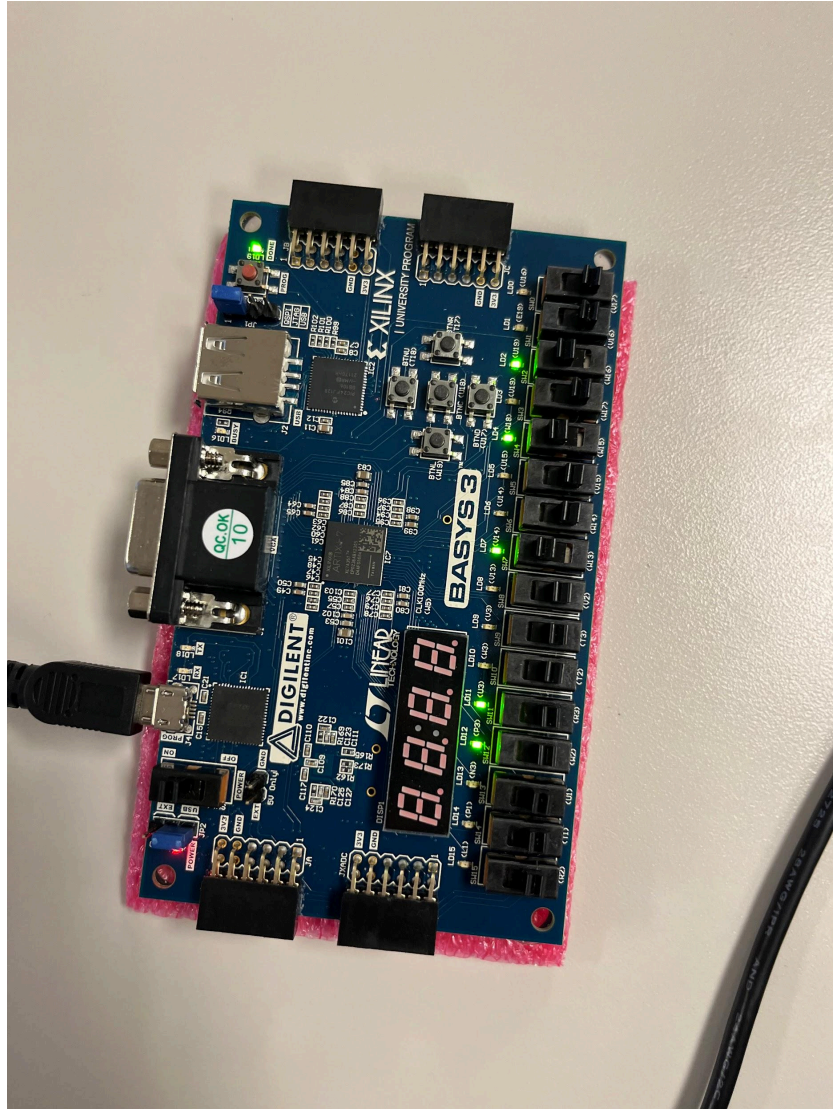
Case 2: The output is expected as the output is 000, zero = 1, carryout = 0, and overflow = 0.



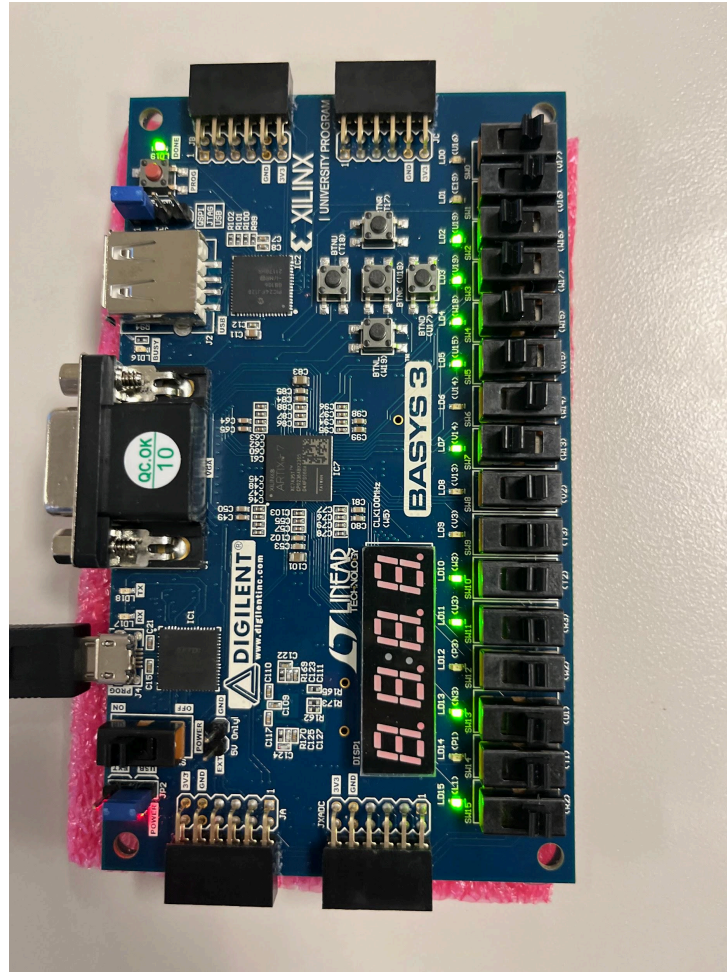
Case 3: The output is as expected as the output is 111, Zero = 0, carryout = 0, and overflow = 0.



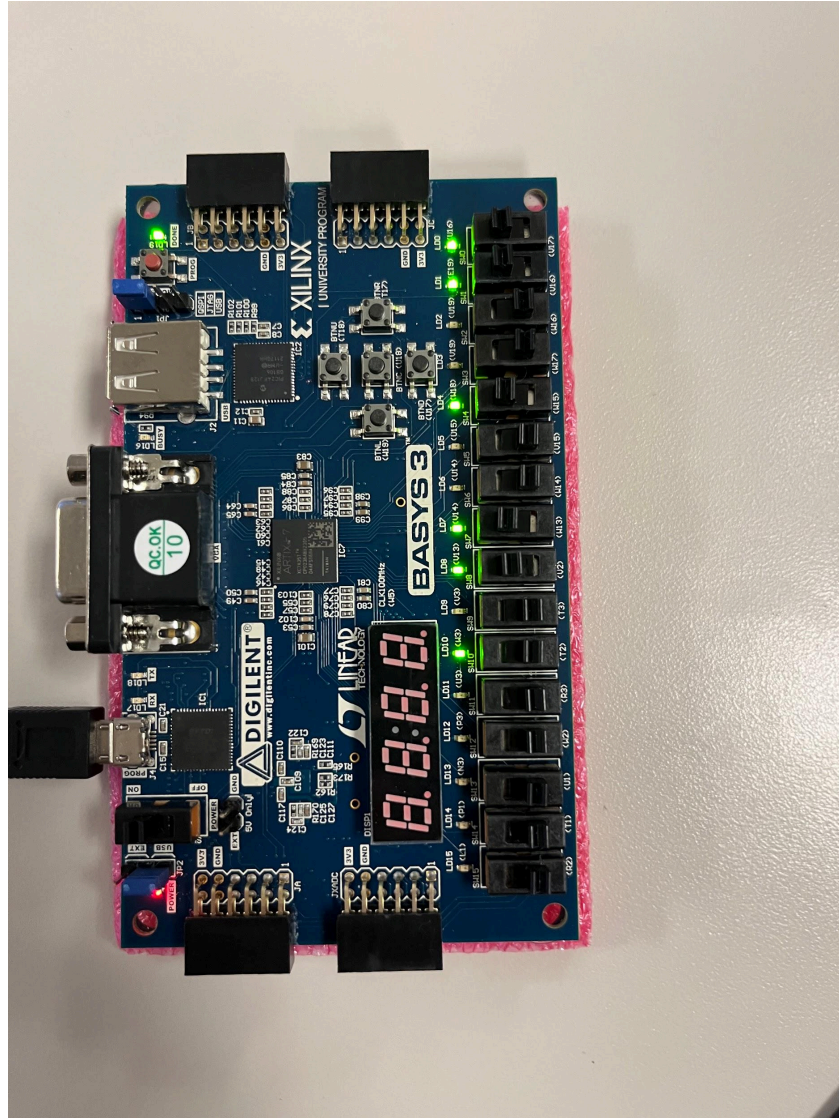
Case 4: The output is as expected as output is 000, Zero = 1, Carryout = 0, and overflow = 0.



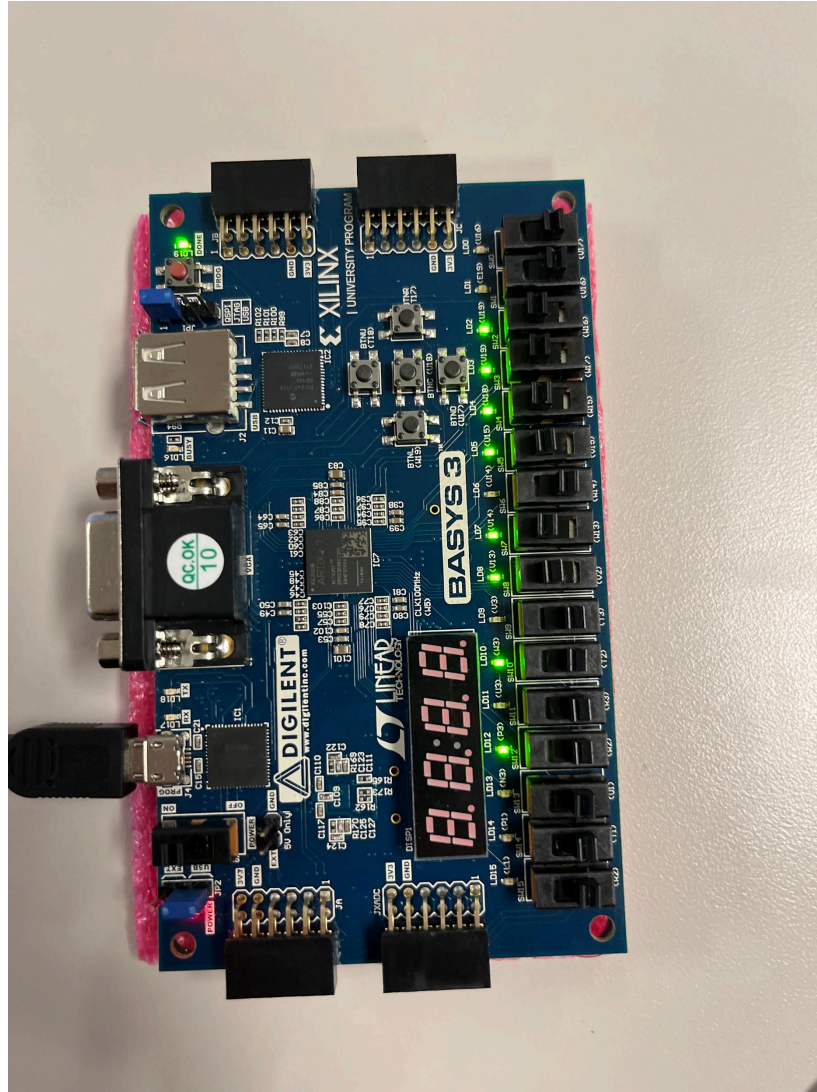
Case 5: The output is as expected as output = 011, zero = 0, carryout = 0, and overflow = 0.



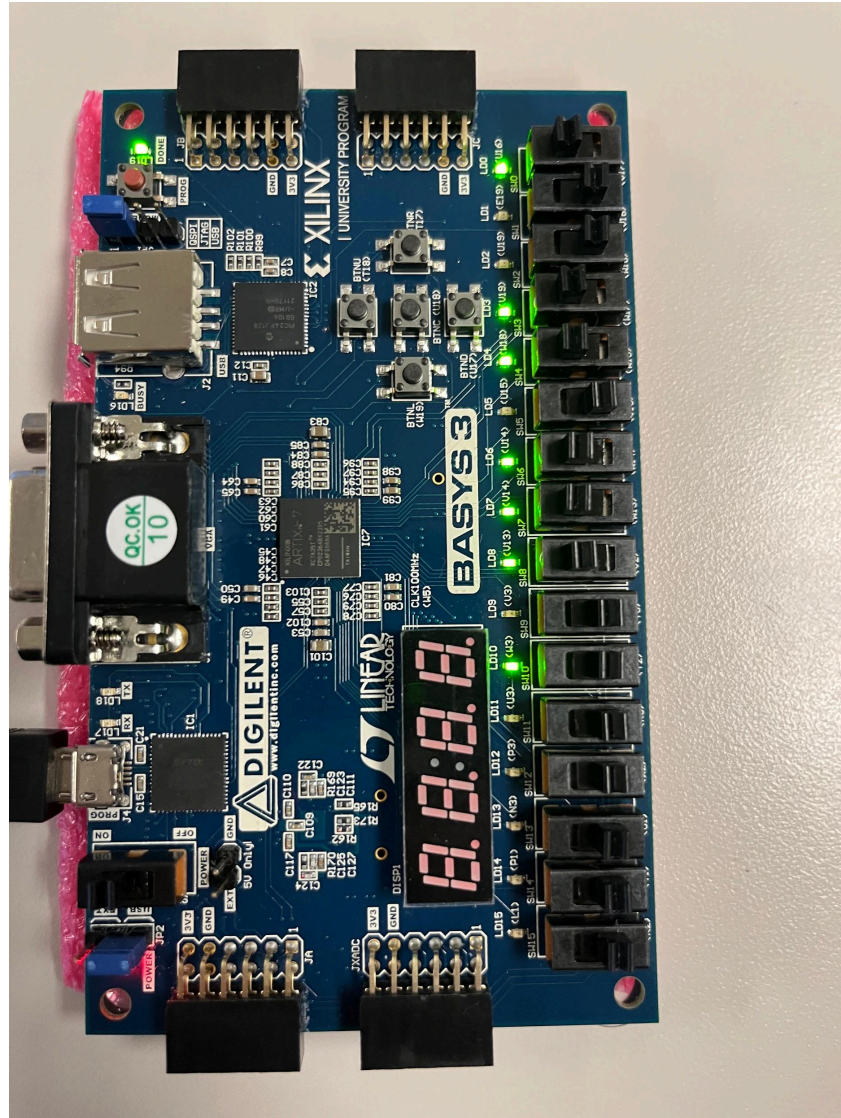
Case 6: The output is as expected as output = 011, zero = 0, carryout = 1, and overflow = 1.



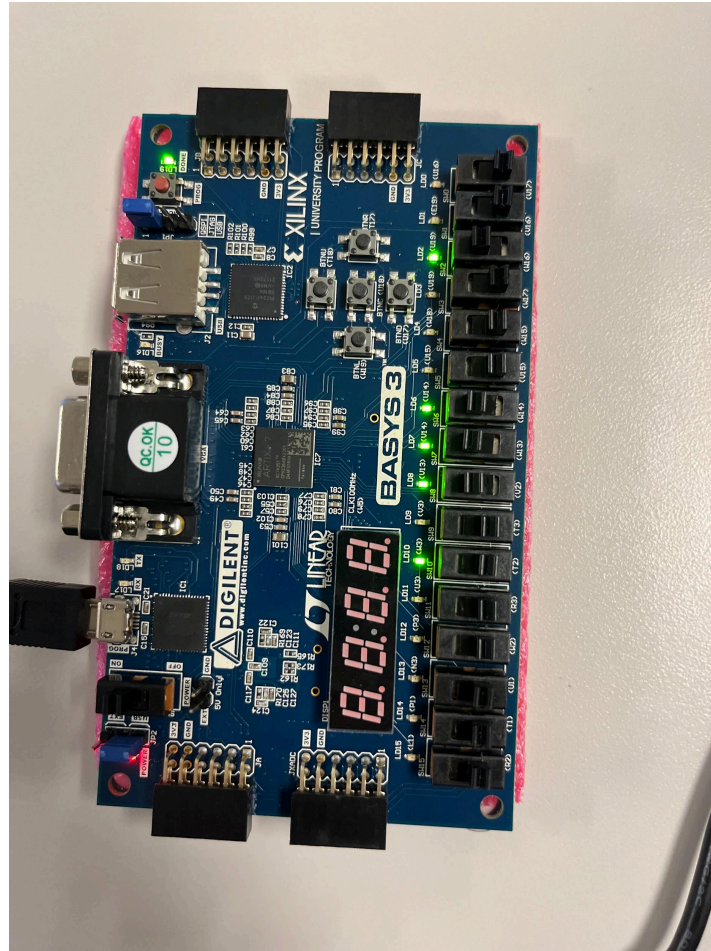
Case 8: The output is as expected as output = 001, zero = 0, carryout = 0, and overflow = 0.



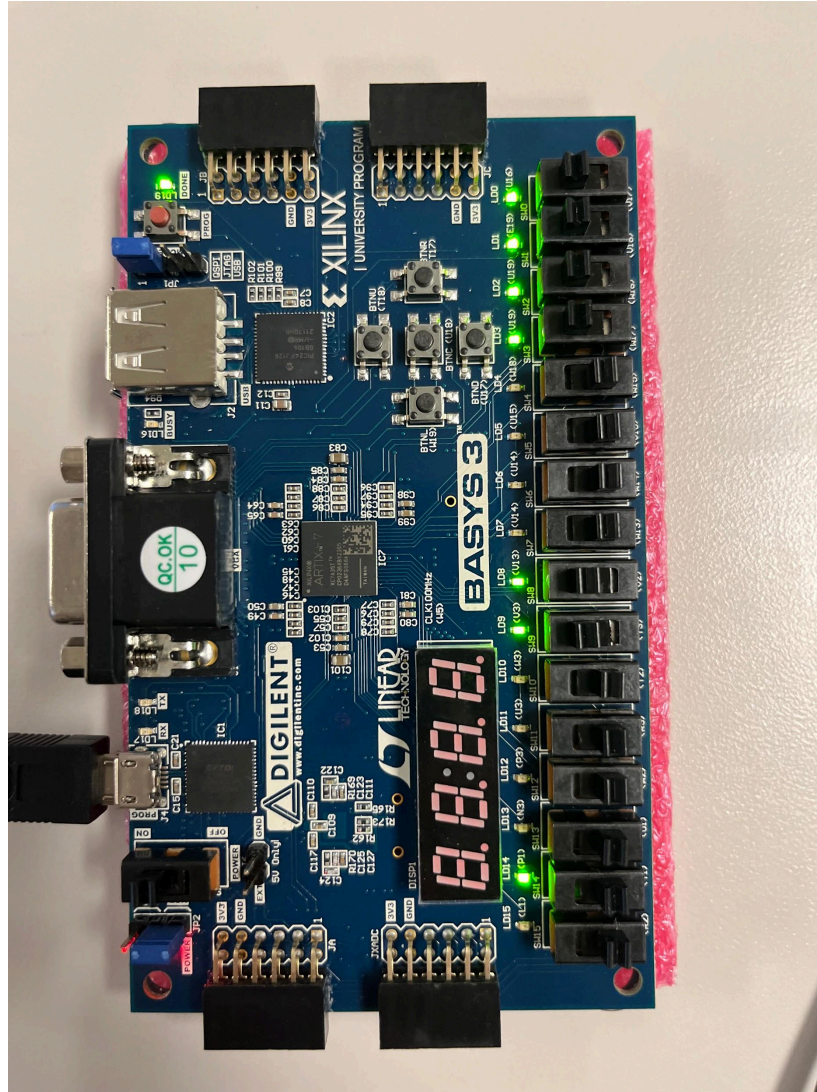
Case 9: The output is as expected as output = 101, zero = 0, carryout = 0, and overflow = 0.



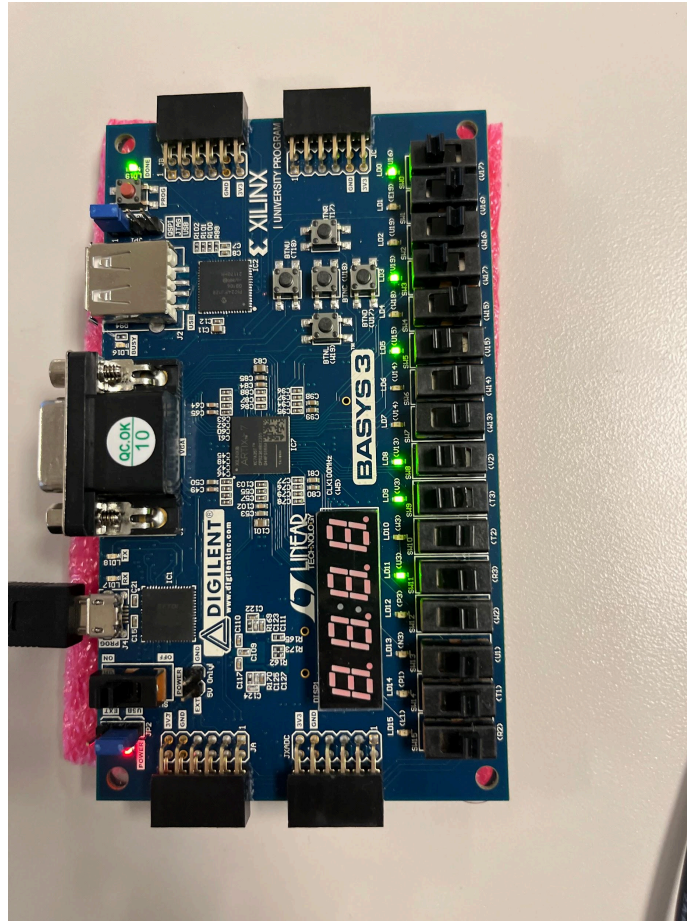
Case 10: The output is as expected as output = 001, zero = 0, carryout = 0, and overflow = 0.



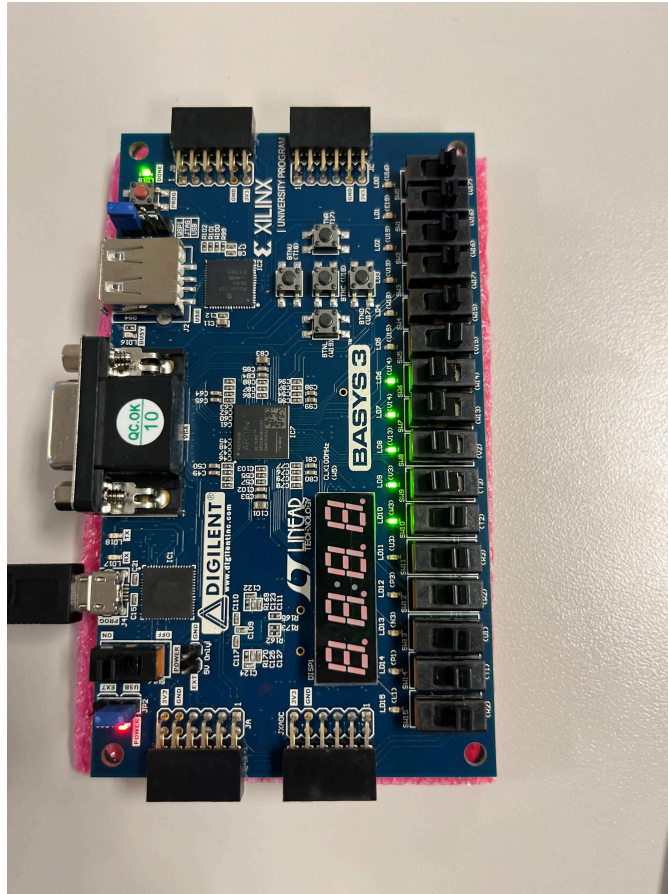
Case 11: The output is as expected as output = 001, zero = 0, carryout = 0, and overflow = 0.



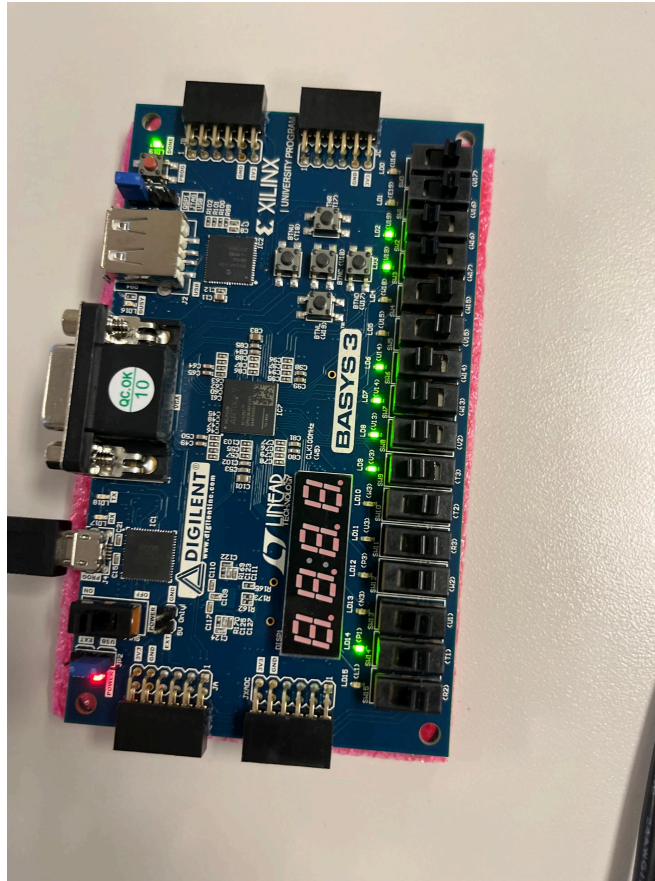
Case 12: The output is as expected as output = 000, zero = 1, carryout = 0, and overflow = 0.



Case 13: The output is as expected as output = 010, zero = 0, carryout = 0, and overflow = 0.



Case 14: The output is as expected as output = 001, zero = 0, carryout = 0, and overflow = 0.



Case 15: The output is as expected as output = 000, zero = 1, carryout = 0, and overflow = 0.